

Use of *Pseudomonas fluorescens* at different concentration to degrade *Amillariella mellea*

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Abstract

Armillaria mellea, known for its root decay illness over different plant species, poses a critical danger to rural efficiency. Conventional control strategies have been ineffective, requiring investigation and exploration into arrangements. *Pseudomonas fluorescens*, known for its different metabolic capabilities and potential as a biocontrol specialist, develops as a promising candidate. This research study investigates the interactions between *Armillaria mellea* fungi and *Pseudomonas fluorescens* bacteria when cultured together on an agar plate. The hypothesis states that changed concentrations of *P. fluorescens* will influence the of *A. mellea* by lowering fungal colony size. To explore this, agar plates were arranged with *A. mellea* and treated with distinctive concentrations of *P. fluorescens*. Picture examination was conducted some time recently and after treatment to evaluate contagious development. The discoveries show that higher concentrations of *P. fluorescens* resulted in a noteworthy decrease in parasitic colonies, illustrating its potential as a biocontrol operator against *A. mellea*. Measurable investigations affirmed the importance of emphasizing the effect of bacterial concentration on parasitic development hindrance. Advance inquiry on delayed bacterial introduction impacts and the relationship between bacterial concentration and time for successful parasitic hindrance is justified. Furthermore, investigating diverse bacterial species and their adequacy against *A. mellea* in different situations may give important bits of knowledge. This thought highlights the potential of *P. fluorescens* as a biocontrol operator against *A. mellea* and underscores the centrality of bacterial concentration in controlling parasitic development. Future examinations sought to dive more profound into optimizing bacterial concentrations and their transient impacts on contagious remediation for viable applications in horticulture. Since so many inferential tests were run the different conclusions about the data are harder to interpret.

Introduction

Armillaria mellea causes an Armillaria root rot disease on a wide range of species including trees, shrubs, and conifer; this disease causes 10–40% annual loss of a common grape vine in California (Devkota & Hammerschmidt, 2020). The fungi is part of facultative necrotrophs, which are known to kill tree roots and feed off of them, thus killing the affected tree (Devkota & Hammerschmidt, 2020). Controlling Armillaria root rot disease (ARR) is challenging and past research seeking solutions have failed or had limited success. A study done by Tampakakis et al. (2009), affiliated with the Boston General Hospital, found *Candida albicans* is a fungal pathogen that affects human health and is commonly found in the human gastrointestinal system. When exposed to blood it can infect the bloodstream causing serious illness (Tampakakis et al., 2009). Biocontrol agents, a possible solution to this issue, are biological organisms that are used to remediate or improve organism life.

Biocontrol is the purposeful introduction of a foreign organism to carry out a specific task (Gao et al., 2012). Biological control (biocontrol) uses many different fields of biology and biotechnology but mostly refers to entomology and plant pathology (Gao et al., 2012). Plant pathology refers to the use of microbial agents to suppress plant disease and use of host specific pathogens to control the population of weeds (Gao et al., 2012). In this process, the organism used to carry out the task is known as a biocontrol agent (BCA) (Gao et al., 2012). According to a study done by Gao et al. (2012), associated with the East China University of Science and Technology, these BCA can be a variety of organisms such as viruses, bacteria, fungi, or plants.

Armillaria mellea is known as a parasite and saprophyte on woody plant hosts; in countries like China and Korea, this mushroom holds medicinal properties (Shim et al., 2006). Shim et al. (2006), from the Department of Biology, Dongguk University, talks about fungal

growing conditions, this fungi is commonly cultured in 18-25°C for the best growth in a moist environment (Shim et al., 2006). A study done by Devkota and Hammerschmidt (2020), from the Michigan State University, found that The fungi also secretes toxins that break down cell walls making the immune system and defense deteriorate by starving the plant (Devkota & Hammerschmidt, 2020).

Pseudomonas fluorescens has versatile metabolic capabilities allowing it to survive in many different environments (Scales et al., 2014). Scales et al. (2014), a researcher for University of Michigan, The bacteria is able to promote plant growth through its antimicrobial mechanisms (Scales et al., 2014). The bacteria can be found in many places including the rhizosphere, soil, and plant surfaces. *P. fluorescens*, from nonmammalian samples, have a growth range from 4-35°C but standard culturing temperature includes a lower range of 5-32°C (Scales et al., 2014). One study done by Gamez et al. (2019), associated with Colombian Corporation of Agricultural Research, used *P. fluorescens* to promote plant growth which shows *P. fluorescens* diverse capabilities and its utilization in research as a biocontrol agent or the use of biotechnology.

Controlling the spread of ARR disease is a challenging obstacle due to its ability to affect a wide-variety of plants Previous studies trying to solve this problem have failed or had limited success, therefore, it is imperative to look into other viable solutions that can limit the growth of this disease so as to keep the fungi from infecting and endangering plant life and health (Devkota & Hammerschmidt, 2020). *P. fluorescens* have been used in the past as a biocontrol agent but limited research is available on its viability to use against fungi. Furthermore, there is a lack of research investigating *P. fluorescens* and *A. mellea* interactions when exposed to each other, and what kind of relationship they will form.

Literature Review

Pesticides and herbicides are compounds that are primarily used in agriculture to keep away unwanted pests, weeds, and diseases (Nicolopoulou-Stamati et al., 2016). A study done by Nicolopoulou-Stamati, related to the Department of Pathology from University of Athens, pesticides and herbicides have been known to cause health issues in humans through touching, inhaling, or ingestions of these substances and with prolonged exposure causing more extreme health issues (Nicolopoulou-Stamati et al., 2016). These substances cause human and animal bodies to bioaccumulate, metabolize, excrete, or store body fat (Nicolopoulou-Stamati et al., 2016). Pesticides and herbicides residues are commonly found in everyday foods and drinks such as cooked meals, wine, water, and other beverages (Nicolopoulou-Stamati et al., 2016). To keep up with worldwide demand, agriculture has turned to pesticides and herbicides as an easy way to increase plant growth. This fact neglects the translocation of these substances up the food chain (Castaldi et al., 2021). As the usage of these compounds continue to increase, health concerns will continue to increase at an alarming rate. Thus explaining why it is important to invest in alternative methods other than pesticides and herbicides to avoid these health concerns.

P. fluorescens, often used as a biocontrol agent for plant pathogens, can also be used as a plant promoting agent (Gao et al., 2012). *P. fluorescens* also reacts differently when it comes across individual organisms and has different remediation capabilities when it comes to different organisms. It is important to investigate the relationship of the organism a study done by Pal & Gardener (2006), from Department of Plant Pathology at the Ohio State University, clearly relationships: the plant and pathogenic host fungi would form a parasitic relationship: while plant and the control agent bacteria would form have a commensalistic interaction where the plant is benefited but the bacteria is neither benefited or harmed: the pathogenic fungi and control agent

bacteria would be in competition for resources and survival. A paper by Rosatto et al. (2021), a researcher in Laboratory of Plant Biology at the University of Genoa, states that the bacterial fungal interaction is important to investigate because it shows how they will interact with each other when used together, what kind of relationship they will make and how that will affect the host.

A study done by Castaldi et al. (2021), affiliated with the Department of Biology at the University of Naples Federico II, researched the use of *P. fluorescens* as biofungicides against *Macrophomina phaseolina*, a pathogenic fungi to soybean plants. This study shows the compounds *P. fluorescens* produces that have antifungal activity and are viable in its use against other kinds of fungi since the paper found the bacteria successful in remediating the fungi (Castaldi et al., 2021). However, this study did not investigate other species of fungi and how the bacteria might respond differently. Also, the solution presented in the paper would have to be looked into more for it to be implemented on a commercial scale (Castaldi et al., 2021).

Fluorescent pseudomonads, which are a group of microorganisms closely related to *P. fluorescens*, are used in the study. The organisms are found to have antifungal activity and they also help increase plant growth (Tampakakis et al., 2009). This research helps relate to how using *P. fluorescens* would be an optimal biocontrol agent because its use as a plant promoter too and how it had antifungal properties backed by many past studies. Although this study does not deal with the same fungi used in this research, it does help give a basic understanding of the interactions that can be inferred from the two (Tampakakis et al., 2009). A research done by Castaldi et al. (2021), affiliated with the Department of Biology at the University of Naples Federico II, this research study used *Bacillus vallismortis*, a bacterial species known for its plant promoting abilities and antifungal activities, to see if it is capable of remediating *Macrophomina*

phaseolina, a pathogenic fungi. The bacteria was successful in remediating the fungi but the study found that it was 10 times less effective than traditional pesticides (Castaldi et al., 2021). This helps show how bacteria can be used as a biocontrol agent for fungi but more research is needed to maximize the effectiveness of treatment so it is as close to or better than pesticides (Castaldi et al., 2021).

This research study tries to investigate the interactions between *Armillaria mellea* fungi and *Pseudomonas fluorescens* bacteria when cultured together on an agar plate. This research hopes to find *P. fluorescens* as a biocontrol agent for the pathogenic fungi and hopes to establish the viability of *P. fluorescens* to remediate *A. mellea*. In addition, investigate how this solution can be commercialized. The independent variable would be the amount of bacteria solution added to the agar plates and the dependent variable would be the size of the fungal culture. It is hypothesized that the highest concentration of *Pseudomonas fluorescens* will have the highest degradation of *Amillariella mellea* since there will be more bacterial colonies present. The experiment will also include a control group to see how big the actual colony would have been without the bacterial interaction.

Materials

Autoclave	Nutrient Broth
Incubator	Distilled Water
Petridish	Flask 200 and 1000 mL
<i>Amillariella mellea</i> culture plate	Bunsen Burner
<i>Pseudomonas fluorescens</i> slant culture	Phone Camera
Nutrient Agar	MAT Labs
Inoculating Loop	Cardboard Boxes
Micropipettes (20-100 μ L)	Paper Napkins
Plastic Straw	Bleach

Methods

First, to set off from the procedure of the research, the agar plates were made. This was made by adding nutrient agar to distilled water: 600 mL of distilled water was mixed with 13.8 g of nutrient agar in two 1000 mL flasks and stirred using a stirring rod. Both flasks were then placed in an incubator. Then they were taken out and poured into 90 petri dishes in addition to 5 extra ones. Then the fungal culture was added to each petri dish; this was done by making plugs by punching holes using a straw from the *Amillariella mellea* culture plate using a plastic straw and adding one into each dish. These petri dishes were cultured at 27°C for 3 weeks, based on a study that had cultured the fungi, in the incubator (Shim et al., 2006). The petri dishes were also numbered from 1-90 and E1-E5 for the 5 additional dishes; the dishes labeled 1-30 were the control group, 31-60 was 100µL bacterial solution, and 61-90 was 200 µL.

The bacterial solution that was added to the petri dishes was made from a past study cultivated bacteria in water by first mixing nutrient broth powder with 200 mL distilled water in a 600 mL flask (Han, 2021). Then the solution was autoclaved to sterilize the solution. The bacteria slant was added to the solution by first adding some distilled water in the slant and adding it to the solution. The agar was broken up and a small piece was added to the solution. The solution was placed in the incubator at 27°C for 24 hours. Then solution was inoculated onto the plates labeled 31-60 100µL, and for plates 61-90, they were inoculated. By first adding the bacterial solution and then streaking the solution with an inoculation loop which was sterilized using a bunsen burner through each trial. The pipette tips were changed when changing from each group. The plates were placed in the incubator for 2 weeks at 27°C. Then pictures of each plate were taken using an Iphone 12 back camera. To get accurate pictures without the glare a set up was made to keep things constant; the phone was laid down flat on two cardboard boxes with

camera exposed and another one on the other side and the sides were covered with paper napkins to avoid the glare on the petri dish. The images were then analyzed on MATLAB. The Image was analyzed by first turning it into an 8-bit image then the gray value was ranked from 0-1 and then threshold values were set so only the pixels with gray value numbers higher than that were added up. The before and after images were tested at different thresholds. Each image was analyzed before and after the experiment to compare the area. The plates were then submerged in bleach to kill any organisms on the plate before disposing of them. A Shapiro Wilk Normality test was done and then a Kruskal-Wallis pairwise comparison was done as well as a Wilcoxon Sign rank test.

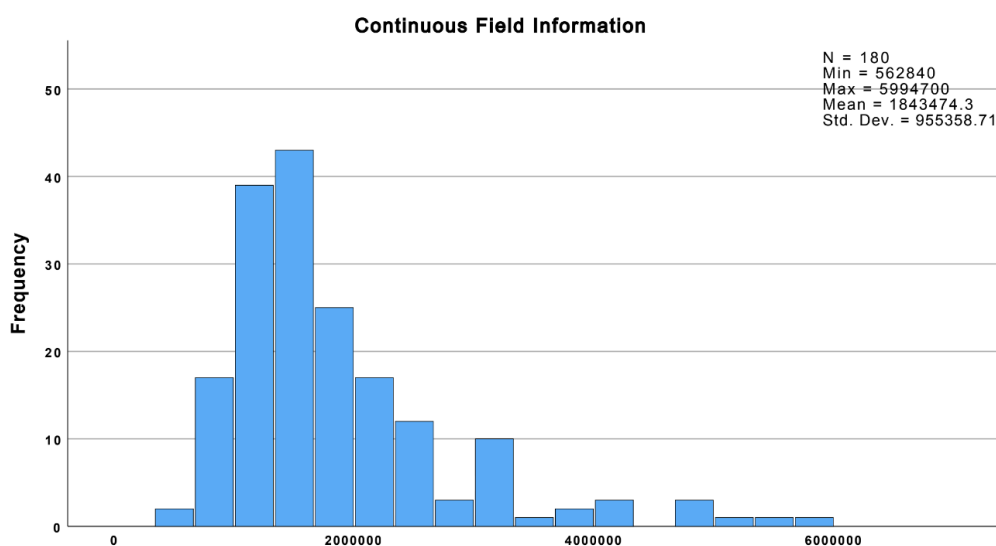
AP Research Proposal EDD

Title of the Experiment Use of <i>Pseudomonas fluorescens</i> at different concentration to degrade <i>Amillariella mellea</i>			
Hypothesis The highest concentration of <i>Pseudomonas fluorescens</i> will have the highest degradation of <i>Amillariella mellea</i>			
Independent Variable Concentration of <i>Pseudomonas fluorescens</i>			
Levels of Independent Variable	Control no <i>Pseudomonas fluorescens</i>	100μL Bacterial Solution	200μL Bacterial Solution
Number of Repeated Trials	30	30	30
Dependent Variable Fungal growth measured by the colony diameter or area.			
Control Group It will not have the bacteria in it, it will only have the fungi.			
Constants			

Results

The data from MATLAB was analyzed from different thresholds. The threshold was shortened to 0.3 to 0.5 instead of 0.3 to 0.8 since the extended threshold range did not encapsulate the data accurately. The thresholds were analyzed separately through different inferential statistics tests.

Histogram Distribution of data in 0.4 Threshold



This shows the distribution of the results at 0.4 threshold. This histogram shows the data that MATLAB gave at 0.4 thresholds. With the number of points, minimum, max, mean, and standard deviation on the top right.

This histogram helps show the distribution of the data the other thresholds were similarly distributed. This helps prove a normality test was needed to ensure that the data was normally distributed. The Shapiro Wilk Normality test found that none of the data through all the thresholds were normally distributed.

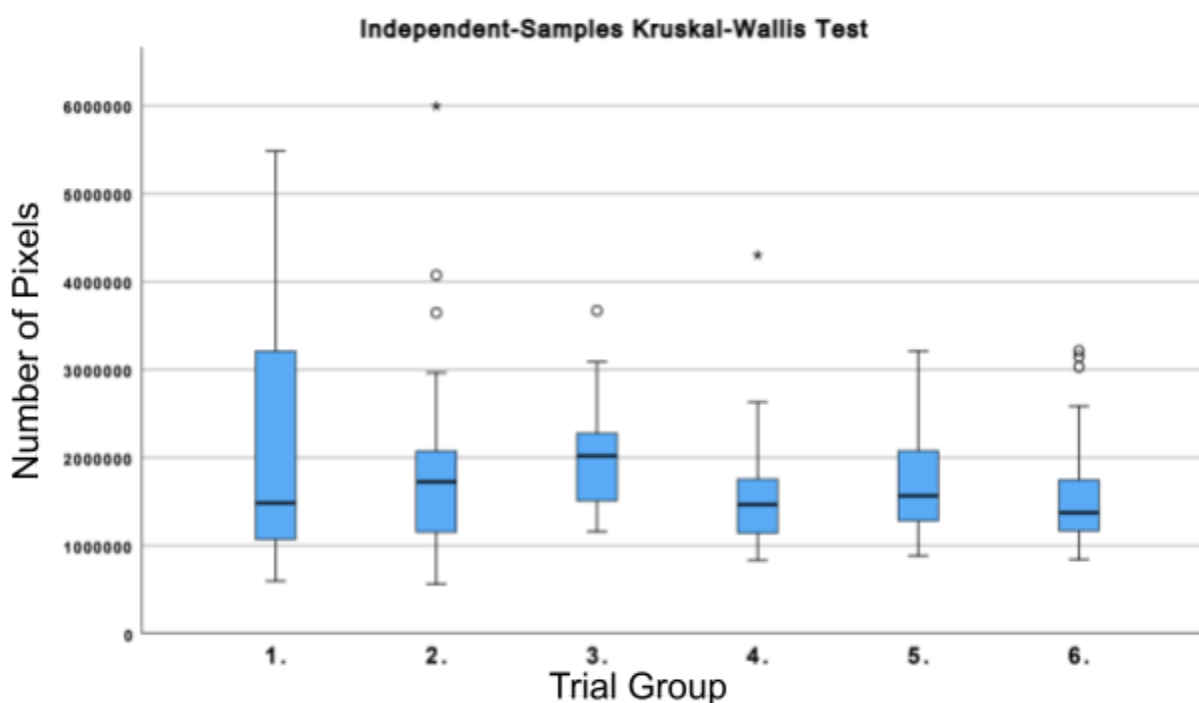
Table Means of Each Group in all Thresholds

Before			After		
1	2	3	1	2	3
2.42133E+06	2.13883E+06	2.11831E+06	1.72727E+06	1.82703E+06	1.76394E+06

Note: This diagram shows the Average values through the shortened threshold of the number of pixels added. The numbers are related to the experimental group with the control group being 1, the group with 100µl bacterial solution being 2, the group with 200µl bacterial solution being 3.

These are the average values of the fungal colony measured through the different thresholds. All the after values are lower than the before values even the control group where no bacterial solution was added is lower than before.

Box-Whisker Plot of Data in 0.4 Threshold



This is the box and whiskers plot that the Kruskal-Wallis Test produced. The numbers are related to the experimental group with 1 being the control group before and 4 being after. 2 being 100µl bacterial solution before and 5 being after. 3 being 200µl bacterial solution before and 6 being after.

This box plot shows the distribution of the trials in before and after analysis. This box plot was taken from the 0.40 threshold which was in the middle of the threshold range which was 0.30-0.50. In the box plot it can be seen that the after trials 4-6 groups had a lower distribution

than the before trials 1-3. But group 1 the control group of before had the greatest distribution on average more than all the other groups. A similar distribution is seen in through the other thresholds.

Table Wilcoxon Sign Rank test

0.3				0.35		
Result 1	Result 2	Result 3		Result 1	Result 2	Result 3
Significant	Significant	Significant		FALSE	FALSE	Significant
0.4				0.45		
Result 1	Result 2	Result 3		Result 1	Result 2	Result 3
FALSE	FALSE	Significant		Significant	FALSE	Significant
		0.5				
		Result 1	Result 2	Result 3		
		Significant	FALSE	Significant		

This table shows the significance of the data shown by running a Wilcoxon Sign Rank test. The tables represent the significance at different threshold values going from 0.3 to 0.5. The numbers headings are related to the experimental group with the control group being 1, the group with 100µl bacterial solution being 2, the group with 200µl bacterial solution being 3.

The Wilcoxon Signed Rank test shows the significance of values when comparing a before and after value. The before and after values were compared in each image. This table shows the significance found in each threshold tested. The test found that the trial tree which was using 200µl had the most significance through multiple thresholds. While 100µl of bacterial solution used found the least significant only being significant at 0.3 threshold. The control group was found to be significant 2 times at 0.3 and 0.5 threshold.

Table of Kruskal Wallis test Before and After Comparison

0.3				0.35		
1	0.011	Significant		1	0.524	Not Significant
2	0.055	Not Significant		2	0.565	Not Significant
3	0.018	Significant		3	0.002	Significant
0.4				0.45		

1	0.184	Not Significant		1	0.091	Not Significant
2	0.231	Not Significant		2	0.565	Not Significant
3	0.003	Significant		3	0.002	Significant
			0.5			
			1	0.091	Not Significant	
			2	0.565	Not Significant	
			3	0.002	Significant	

This shows the results of the Kruskal Wallis test. This test is showing the results of the before and after. The numbers headings are related to the experimental group with the control group being 1, the group with 100µl bacterial solution being 2, the group with 200µl bacterial solution being 3.

This table shows the results of the Kruskal Wallis test. The test compares the before and after values of each group. First a statistical significance was found then a pairwise comparison was done to find out which pairs were significant. This test found group 3 which had 200µl bacterial solutions to have the most significance through different thresholds. While group 2 with 100µl bacterial solution wasn't found to be significant a single time which does not correlate with Wilcoxon Signed Rank where it was significant at 0.3 threshold. This test also only found control group to be significant at 0.3 threshold only where Wilcoxon Signed Rank found it to be significant two times at 0.3 and 0.5.

Table of Kruskal Wallis Pairwise Comparison between After

0.3				0.35		
4-5	0.128	Not Significant		4-5	0.364	Not Significant
4-6	0.208	Not Significant		4-6	0.758	Not Significant
5-6	0.792	Not Significant		5-6	0.549	Not Significant
0.4				0.45		
4-5	0.569	Not Significant		4-5	0.569	Not Significant
4-6	0.617	Not Significant		4-6	0.617	Not Significant
5-6	0.284	Not Significant		5-6	0.284	Not Significant
		0.5				
		4-5		0.569		Not Significant

	4-6	0.617	Not Significant
	5-6	0.284	Not Significant

This table shows the comparison of the after trials to see if there was a statistically significant difference between after pictures by using a Kruskal Wallis test pairwise comparison. The numbers represent what was compared with what 4 is the control group after, 5 is 100µl of bacterial solution after, 6 is 200µl of bacterial solution after.

This table is the Kruskal Wallis test run on the after trials. A significance was not found between the after trials. This test found that there was no statistical difference between after trials through all the thresholds. The test found that even between the control group where there was no bacterial solution added had no statistically significant difference.

Discussion

The purpose of this research was to find *P. fluorescens* as a biocontrol agent for the pathogenic fungi and hopes to establish the viability of *P. fluorescens* to remediate and control the growth of *A. mellea*. In addition, investigate how this solution can be commercialized. It was hypothesized that the 200µl of *Pseudomonas fluorescens* would have the greatest remediation of *Amillariella mellea* since there is the greatest amount of the bacteria to remediate the fungi. This hypothesis was supported since the data collected showed that the 200µl of bacterial solution used did have a statistically significant difference between the before and after trials.

The threshold was shortened from 0.3 to 0.8 to 0.3 to 0.8 because when the images were analyzed with threshold from 0.5 to 0.8 the fungal colonies were not accurately encapsulated which would skew the results. These different thresholds were used to help determine the best fungal colony measurement in each plate since different thresholds can give different results. Using different thresholds it can be ensured that the fungal colonies were measured as accurately as possible.

The Shapiro Wilk Normality test found that none of the data was normally distributed which means non parametric inferential statistics tests were used for this data. A Kruskal Wallis

test and Wilcoxon Sign rank test were used which were the best test for this kind of non-parametric test with before and after data. First the box plot shows how only the control group before was the highest with trial 1 and trial 2 before right behind. But the surprising thing is the control group after was also lower than the before. Since there was no bacterial solution added the fungal colony should have grown and increased in size. But both inferential statistics found that that difference was not statistically significant. This might have been due to multiple reasons such as depletion of resources which might have happened but the fungal colonies did not cover all the agar which would not make sense. Another cause could have been because there was growth of the fungal colony but it was in the z-axis where it grew upwards and the pictures taken did not have the z-axis in them which could be a possible explanation.

Then looking at the inferential test run on the data it can be seen that trial 3 with 200 μ l bacterial solution had the greatest significance. Both Kruskal Wallis and Wilcoxon Sign rank tests found trial 3 to be statistically significant through all the thresholds. While trial 2 with 100 μ l of bacterial solution was not significant through the Kruskal Wallis test but was significant one time in Wilcoxon Sign rank test at threshold 0.3. Since this trial was only found to be significant one time at one threshold it can be safe to assume that this trial was not as effective as trial 3. But this does not explain why the Kruskal Wallis test and Wilcoxon Sign rank test found different results at 0.3 threshold this can be because of how each test works since Wilcoxon Sign rank test is used for data with two sets such as before and after while Kruskal Wallis test is similar to a one-way ANOVA but the pairwise comparison should have resulted in similar results. The control group was also only found to be statistically significant in the 0.3 in the Kruskal Wallis test and 2 times in the Wilcoxon Sign rank test which does not entirely support there was a statistically significant difference but there could have been. The data helps support

that the 200µl trial did have an impact on the fungal colonies in which they had decreased in size as a result of the bacterial solution. While 100µl did not prove to have the same effect. As a result it led to believe that a higher concentration of bacterial solution could help the bacteria remediate the fungi faster and be more effective. A study done by Ayilara and Babalola (2023) talks about what microorganisms like bacteria do when they convert toxic chemicals into less toxic materials which helps explain why a higher concentration of these organisms can help since there is a higher concentration of bacteria for it to counteract the fungal growth. A study in the past also found that the bacteria can be effective biofungicides against *Macrophomina phaseolina*, a pathogenic fungus. The present study helps support this one since it was also successful at remediating the fungi at a higher concentration.

The results of this study help shed new light on the possible uses of *P. fluorescens* in nature. The bacteria is already known for its ability to promote plant growth. But with this new information it helps determine other possible impacts of this bacteria like its ability to control and remediate pathogenic fungi. This helps learn about the impacts of commercializing the bacteria and implementing it in nature. This also helps find how the bacteria can be less effective and more costly than traditional herbicides.

Some more research would have to be done to investigate the potential impact of other bacteria already in the soil and how it would compete or help them. More research would also need to be done on how effectiveness can be improved and less costly than herbicides. Also different methods can be commercialized and implemented in nature. A possible implementation can be inoculating the bacteria in water and watering the infected areas similar to how in this study the bacteria was added to the plates by adding the bacterial broth.

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